# Voice Agentic Smart Reader Application In Sinhala and Tamil For Visually Impaired People

***Abstract*—Access to documents, news articles, and printed content in low-resource languages such as Sinhala and Tamil remains a significant challenge for visually impaired and reading-challenged users. While recent advances in computer vision, speech processing, and large language models (LLMs) have enabled powerful document understanding capabilities, existing systems often operate in isolation, providing limited support for non-visual navigation, emotional expressiveness, and interactive knowledge access. This research presents a multi-component intelligent smart reading system that integrates document acquisition, structured understanding, interactive retrieval, and emotionally expressive speech synthesis for Sinhala and Tamil languages. The proposed system consists of four tightly coupled modules. First, an Intelligent VoiceGuided Navigation (IVGN) component enables non-visual document capture by transforming real-time geometric information from a semantic segmentation backbone into adaptive, multilingual auditory guidance, ensuring stable alignment and reliable auto-capture of OCR-ready images. Second, a Tamil article understanding and question answering module performs semantic paragraph clustering and safe heading generation, followed by an article-grounded Retrieval-Augmented Generation (RAG) pipeline with controlled hybrid web fallback to support accurate, context-aware querying. Additionally, the module tracks user reading behavior to recommend related news articles retrieved from the internet and uses reinforcement learning to continuously improve personalization. Third, a voice-assisted Sinhala smart reader enables non-linear document navigation through Sinhala voice command classification, semantic segmentation, and vector-based retrieval, allowing users to interact with documents using natural spoken instructions. Finally, an Emotional Sinhala and Tamil Text-to-Speech (TTS) component generates emotionally expressive speech using emotion-conditioned acoustic modeling and controllable emotion intensity, enhancing the naturalness and communicative effectiveness of auditory output under limited data conditions. Experimental evaluations across components demonstrate reliable document capture, effective topic segmentation, accurate document-grounded question answering, improved navigation efficiency, and emotionally distinguishable speech synthesis. Together, these contributions form a unified, accessible, and human-centric smart reading framework that advances intelligent document interaction for underrepresented languages.***

***Index Terms*—Smart reading systems; low-resource languages; Sinhala and Tamil; assistive technologies; intelligent document acquisition; voice-guided navigation; semantic segmentation; optical character recognition (OCR); retrieval-augmented generation (RAG); document-grounded question answering; voice-assisted navigation; emotional text-to-speech; human–computer interaction; accessibility.**

## I. Introduction

Access to textual information is essential for education, employment, and independent living. However, visually impaired and reading-challenged users continue to face significant barriers when inter- 1 acting with printed and digital documents, particularly in low-resource language contexts such as Sinhala and Tamil. Although advances in computer vision, speech processing, and natural language understanding have enabled powerful assistive technologies, existing systems often address individual sub-tasks in isolation, resulting in fragmented user experiences that lack seamless interaction, adaptability, and accessibility. Mobile document acquisition systems have substantially improved access to printed content through advances in optical character recognition (OCR) [1], [4]. Nevertheless, accurate OCR performance remains highly dependent on proper geometric alignment, scale consistency, and illumination during image capture. For visually impaired users, achieving such alignment without visual feedback is a major challenge. Most existing document scanning applications rely on visual overlays or static audio prompts, which provide limited contextual awareness and fail to adapt to document orientation or user behavior. As a result, users experience repeated corrections, prolonged capture times, and reduced usability during document acquisition. Beyond capture, interacting with extracted text presents additional challenges. Conventional screen readers typically operate using linear reading paradigms, forcing users to listen sequentially rather than allowing efficient navigation based on intent. This limitation is particularly pronounced in low-resource languages such as Sinhala, where intelligent voice-driven document navigation systems remain underdeveloped. Sinhala is morphologically rich and permits multiple linguistic variations for the same semantic command, complicating robust voice command recognition and intent mapping. As a result, users lack the ability to naturally navigate documents, request summaries, or access specific content sections through spoken interaction. Recent advancements in Large Language Models (LLMs) have significantly improved text understanding, summarization, and question answering in high-resource languages [7], [11], [12], [15]. However, similar capabilities for low-resource languages such as Tamil remain limited, especially in structured news article understanding and document-grounded question answering. Tamil news articles often contain multiple subtopics distributed across paragraphs without explicit structural boundaries, making topic-aware summarization and focused querying difficult. Direct application of LLM-based generation without grounding further risks hallucinated or unreliable responses. Retrieval-Augmented Generation (RAG) frameworks mitigate this issue by grounding generation in retrieved context [8], [19], yet their application to structured Tamil article

understanding and adaptive retrieval remains underexplored. In addition to navigation and understanding, speech output quality plays a critical role in assistive systems. Text-to-Speech (TTS) technology serves as the primary interface through which visually impaired users access documents, summaries, and interactive responses. While modern neural TTS systems achieve high intelligibility in high-resource languages, TTS solutions for Sinhala and Tamil largely produce neutral and monotonous speech. The lack of emotional expressiveness reduces user engagement, comprehension, and the perceived naturalness of interaction, particularly in long-form reading and assistive feedback scenarios. Emotional speech synthesis in low-resource languages is further constrained by limited datasets, speaker imbalance, and recording artifacts. To address these challenges holistically, this research presents a multi-component intelligent smart reading system designed for low-resource languages. The system integrates four complementary components: (1) an Intelligent Voice-Guided Navigation (IVGN) module that enables nonvisual document capture by transforming real-time geometric information from semantic segmentation into adaptive, multilingual auditory guidance; (2) a voice-assisted Sinhala smart reader that supports non-linear document navigation through natural Sinhala voice command classification, semantic segmentation, and retrieval-augmented summarization; (3) a Tamil article understanding and question answering module that performs semantic topic segmentation, safe heading generation, and document-grounded RAG with controlled hybrid web fallback; and (4) an Emotional 2 Sinhala and Tamil Text-to-Speech component that generates emotionally expressive speech using emotion-conditioned acoustic modeling with controllable intensity under limited data conditions. By unifying intelligent document acquisition, structured understanding, interactive retrieval, and emotionally expressive speech synthesis, the proposed system advances toward a more accessible, adaptive, and human-centric smart reading framework. This work contributes practical solutions for intelligent document interaction in underrepresented language environments and demonstrates the feasibility of building end-to-end assistive reading systems tailored to the needs of visually impaired users.

## II. Related Work

Prior work relevant to this study spans OCR-enabled document acquisition, non-visual guidance, multilingual NLP, retrieval-augmented generation, and emotional speech synthesis. Transformer-based OCR and document quality assessment methods have improved text extraction robustness, while assistive guidance studies show that adaptive auditory feedback can significantly reduce user burden during capture. In low-resource settings, recent multilingual transformer models and parameter-efficient adaptation methods have improved classification, summarization, and question-answering performance, but robust Sinhala/Tamil integration remains limited. Existing screen-reader workflows are still largely linear and neutral in speech delivery, highlighting the need for a unified system that combines non-visual capture, structured retrieval, voice-based navigation, and expressive TTS in one end-to-end architecture.

## III. Methodology

This section describes the technical methodology used to implement the proposed multilingual intelligent smart reading system. The system integrates four core components: (1) Intelligent VoiceGuided Navigation (IVGN) for non-visual document capture, (2) Tamil article understanding and retrieval-based question answering, (3) Sinhala voice-assisted non-linear document navigation, and (4) Emotional Sinhala and Tamil Text-to-Speech (TTS). The overall methodology is designed to ensure accessibility, structured understanding, reliable retrieval, and natural voice interaction under low-resource language constraints.

### A. System Architecture Overview

The proposed system follows a modular, multi-component architecture designed to provide accessible and intelligent document interaction for visually impaired users in low-resource languages such as Sinhala and Tamil. the architecture integrates several independent yet interoperable modules that collectively handle document acquisition, text extraction, semantic understanding, voice-based interaction, and emotionally expressive speech output. The system workflow begins with document capture using a mobile device, where an external module performs image preprocessing and optical character recognition (OCR) to extract textual content. The extracted text is then processed through language identification and routed to language-specific processing pipelines. Tamil text is directed to a structured article understanding and question answering module that employs semantic segmentation and retrieval-augmented generation for accurate information retrieval, while Sinhala text is indexed for voice-assisted navigation that allows users to interact with documents through natural voice commands. The processed results—including summaries, retrieved sections, or answers—are subsequently passed to an emotional text-to-speech synthesis module that generates expressive speech output in Sinhala or Tamil. The architecture adopts a loosely coupled design, allowing each module to operate independently while maintaining seamless integration across the overall pipeline. This modular approach improves scalability, facilitates component upgrades, and ensures reliable performance under low-resource conditions, ultimately enabling a human-centered smart reading framework tailored to the needs of visually impaired users.

### B. Datasets

This research employs a comprehensive combination of publicly available datasets and manually curated language resources to support emotion classification, emotional text-to-speech synthesis, voice-assisted non-linear document navigation, and robust document understanding for Sinhala and Tamil. Given the low-resource nature of both languages and the real-world deployment context for visually impaired users, particular emphasis was placed on selecting datasets that

reflect authentic linguistic usage, emotional expression, and realistic document capture conditions while ensuring ethical use, scalability, and reproducibility. Emotion Classification Datasets For text-based emotion classification, the study utilizes the ACTSEA (AISECTA) Annotated Corpus for Tamil and Sinhala Emotion Analysis, a large-scale dataset constructed from real-world Twitter data. The raw corpus contains approximately 600,280 Tamil tweets and 318,308 Sinhala tweets, collected through a semi-automatic pipeline that combines emotion seed keywords, morphological expansion, and large-scale Twitter scraping. From this raw collection, a gold-standard subset was created through manual validation by native Sinhala and Tamil annotators [14], resulting in 1,800 Tamil samples (300 per emotion) and 1,200 Sinhala samples (200 per emotion). The dataset covers six basic emotions—anger, disgust, fear, joy, sadness, and surprise—aligned with Ekman's basic emotion model. The corpus explicitly accounts for the morphological richness, informal expressions, and cultural nuances of both languages, enabling realistic emotion analysis in low-resource settings. Emotional Speech Dataset for Sinhala TTS To support emotion-aware speech synthesis, a manually collected Sinhala emotional speech dataset was created specifically for this research. The dataset consists of 500 audio recordings [17], evenly distributed across five emotion categories: happy, sadness, anger, fear, and neutral. All recordings were produced by a single speaker in a controlled, acoustically stable environment to ensure voice consistency, which is critical for emotional text-to-speech modeling. The dataset contains a total of 500 audio recordings, created by first selecting 100 neutral Sinhala sentences and then recording each sentence across five emotion categories: happy, sadness, anger, fear, and neutral, resulting in five recordings per sentence. The speaker was instructed to maintain appropriate emotional prosody, pitch variation, and speaking intensity for each category. To ensure the accuracy and naturalness of emotional expression, each recorded audio sample was evaluated by a professional drama academy instructor. In addition, video recordings of the emotional performances were conducted with the assistance of a drama student, enabling visual verification of emotional delivery during the recording process. This dataset directly addresses the lack of publicly available emotional speech resources for Sinhala and forms a core contribution of the study. Sinhala and Tamil Text Corpora for Language Understanding To support semantic understanding, document segmentation, and contextual processing, largescale textual datasets were incorporated. The Navanjana Sinhala Articles dataset, containing over 2.1 million Sinhala text passages, was used to analyze linguistic structure, semantic segmentation behavior, and long-form document characteristics. The corpus includes news articles, Wikipedia entries, and general web content, providing diverse writing styles and natural topic transitions commonly encountered in real-world reading materials. Although the dataset lacks explicit topic or summary annotations, its scale and diversity make it suitable for semantic segmentation, retrievalaugmented summarization, and document-level understanding in Sinhala. For Tamil language modeling and summarization tasks, a public Tamil news dataset consisting of approximately 20,000 structured news records was utilized. To simulate realistic newspaper-style documents, five consecutive news articles were aggregated into a single training sample, resulting in approximately 4,000 multi-topic documents. This aggregation strategy increases contextual depth and enables modeling of long-range dependencies, supporting multi-topic summarization and context-aware question answering for Tamil. Sinhala Command Dataset for Voice-Assisted Navigation To enable voice-assisted non-linear document navigation, a custom Sinhala command dataset was manually constructed to train and evaluate the command classification model. The dataset consists of approximately 2,000 Sinhala command samples categorized into 15 predefined command classes, including navigation and interaction intents such as NEXT, PREVIOUS, REPEAT, SUMMARIZE, and GO_TO_SECTION. The dataset captures the linguistic variability of Sinhala commands, where multiple syntactic forms can express the same semantic intent. Manual curation ensured linguistic correctness, diversity, and natural interaction patterns, enabling supervised training of robust command classification models capable of generalizing across varied user expressions. Dataset Selection Strategy for Document Segmentation The dataset selection strategy for document segmentation was guided by the need to ensure robustness under real-world document capture conditions, including variable illumination, complex backgrounds, perspective distortion, geometric deformation, and partial occlusion. Rather than relying on synthetic or scanner-based datasets, publicly available camera-captured document datasets were selected to better reflect real usage scenarios encountered in mobile and handheld capture by visually impaired users. Two complementary datasets were used for training and evaluation of the document segmentation model [5], [6]. Document Segmentation v2 Dataset (Roboflow Universe) The Document Segmentation v2 dataset from Roboflow Universe was used as the primary dataset for learning document boundary localization. The dataset contains 1,763 camera-captured document images, annotated using COCO-format polygon segmentation masks representing document regions at pixel-level precision. All images were resized to $640 \times 640$, with automatic orientation correction applied through EXIF metadata stripping. No artificial data augmentation was applied. The dataset includes documents captured under diverse real-world conditions, such as varying lighting levels, heterogeneous backgrounds, scale variations, and orientation changes. The use of polygon-based segmentation masks enables precise contour prediction, which is critical for stable quadrilateral extraction and reliable voice-guided navigation. The absence of synthetic augmentation ensures that robustness is derived from inherent dataset diversity rather than artificially 11 induced transformations. WarpDoc Dataset (Visual Intelligence Lab) To improve robustness against geometric deformation and perspective distortion, the WarpDoc dataset from the Visual Intelligence Lab was incorporated. This dataset consists of 1,020 images collected from sources

such as scientific papers, magazines, and envelopes, captured in both indoor and outdoor environments. The dataset includes multiple structured distortion classes, including rotated, folded, curved, perspective-transformed, randomly deformed, and incomplete document views. WarpDoc introduces deformation patterns commonly encountered in handheld mobile capture scenarios, such as non-planar bending, oblique camera angles, and partial visibility. Training with deformation-rich samples enhances model generalization and reduces failure cases in real-time segmentation. From a geometric perspective, the diversity of distortions expands the distribution of possible quadrilateral configurations, enabling the segmentation model to learn invariance across perspective projections and geometric perturbations. Combined Dataset Impact The combined document segmentation dataset size of 2,783 images enables robust contour prediction, reduces overfitting to ideal planar documents, and improves the stability of quadrilateral smoothing under dynamic capture conditions.

Emotional Sinhala TTS Dataset

### *C. Integrated System Components*

#### 1) Document Capture and Voice-Guided Navigation System

This module is designed to help visually impaired users capture high-quality document images suitable for Optical Character Recognition (OCR). The system uses an Intelligent Voice-Guided Navigation (IVGN) mechanism that analyzes real-time document geometry in the camera frame and provides adaptive auditory guidance.

A document segmentation model was trained on a combined dataset of 2,783 camera-captured document images spanning variation in illumination, background clutter, and geometric distortion. Performance was measured using Intersection over Union (IoU) between predicted document masks and ground-truth polygon annotations, achieving a validation IoU of 97.48%.

Geometric stability was further evaluated through frame-to-frame analysis of centroid position, scale ratio, and orientation. Temporal smoothing reduced oscillations caused by hand motion and segmentation noise, enabling more stable alignment estimation. The module also includes an automatic capture mechanism that triggers image acquisition only when alignment thresholds remain stable across consecutive frames, reducing premature captures and improving reliability in real-world use.

#### 2) Language Understanding and Retrieval Evaluation (Tamil and Sinhala)

The Tamil language understanding component was evaluated through comparative testing of multiple open-source instruction-tuned language models. Initial experiments revealed issues such as unintended language output, repetitive generation, and hallucinations. A Tamil LLaMA-based quantized model demonstrated the most stable performance and was selected for further experimentation.

Prompt optimization and threshold tuning played a critical role in improving reliability. A similarity threshold of 0.85 provided balanced topic segmentation, while retrieval-threshold enforcement significantly reduced hallucinated responses. Hybrid retrieval experiments confirmed that the system successfully distinguishes between document-grounded and out-of-document queries, activating web-based fallback only when necessary.

#### 3) Sinhala Voice Command Classification and Navigation System

The Sinhala Voice Command Classification and Navigation System enables hands-free interaction by allowing users to control document navigation using spoken Sinhala commands. A custom dataset consisting of approximately 2,000 voice command samples across 15 command classes was collected and used to train several machine learning models, including Support Vector Machines (SVM), Logistic Regression, and Neural Networks. Among these approaches, the SVM classifier demonstrated the most consistent performance, particularly in handling the linguistic variability and morphological complexity of Sinhala language commands. Feature representation experiments indicated that the combination of Term Frequency–Inverse Document Frequency (TF-IDF) and SVM achieved the best balance between classification accuracy and computational efficiency for short voice commands. The system was further evaluated under different navigation scenarios, including linear navigation, repeated command execution, and non-linear command sequences. Results showed that the system accurately updated document states, maintained navigation history, and handled boundary conditions effectively while supporting advanced commands such as summarization and section jumping.

#### 4) Emotional Text-to-Speech (TTS) System

The Emotional Text-to-Speech (TTS) system was developed to generate expressive speech output that improves the listening experience of visually impaired users. Unlike conventional TTS systems, which often produce neutral and monotonous speech, this component incorporates emotion recognition to adapt tone, prosody, and delivery style. Emotion classification was performed separately for Tamil and Sinhala to ensure accurate emotional interpretation across distinct linguistic contexts. For Tamil emotion recognition, the multilingual XLM-RoBERTa (XLM-R) model was used [16], [15], and experiments showed that appropriate learning-rate tuning was essential for stable training and reliable classification performance.

For Sinhala emotional speech analysis, both traditional machine learning and deep learning approaches were evaluated. Results indicated that machine learning-based models achieved classification accuracies of up to 90%, demonstrating strong capability in identifying emotional states from speech features. Predicted emotion labels, such as happiness, sadness, and neutrality, were then integrated into the Emotional TTS pipeline. These labels were used to modulate speech synthesis parameters, enabling generated speech to convey suitable emo-

TABLE I
EMOTION CLASSIFICATION PERFORMANCE COMPARISON

| Language | Model | Classes | Accuracy | Macro F1 |
|---|---|---|---|---|
| Tamil | DistilBERT Multilingual | 6 | 78% | 0.78 |
| Tamil | XLM-RoBERTa | 6 | 85.43% | 0.83 |
| Sinhala | SinBERT | 6 | 67% | 0.66 |
| Sinhala | XLM-RoBERTa | 6 | 87% | 0.87 |

TABLE II
TAMIL EMOTION CLASSIFICATION PERFORMANCE

| Emotion | Precision | Recall | F1-Score |
|---|---|---|---|
| Anger | 0.64 | 0.54 | 0.58 |
| Fear | 0.93 | 0.93 | 0.93 |
| Happy | 0.94 | 0.94 | 0.94 |
| Sadness | 0.74 | 0.81 | 0.78 |
| Surprise | 0.90 | 0.95 | 0.93 |

tional expression.To evaluate synthesis quality, a subjective listening study was conducted. Participants rated synthesized speech samples based on emotional clarity, naturalness, and overall listening comfort. The findings showed that emotion-aware TTS consistently outperformed neutral speech synthesis, particularly in conveying emotions such as happiness and sadness. Overall, these results indicate that incorporating emotional context into speech synthesis substantially improves naturalness, expressiveness, and usability in assistive voice-based applications.

## IV. RESULTS

### A. Emotion Classification Results

The emotion classification component was evaluated for both Tamil and Sinhala using transformer-based multilingual models. Experiments were conducted with DistilBERT, SinBERT, and XLM-RoBERTa architectures. Models were trained on labeled emotion datasets and evaluated using accuracy and macro F1-score.

As shown in Table I, XLM-RoBERTa achieved the best performance for both languages, with 85.43% accuracy for Tamil and 87% for Sinhala. This performance gain is attributed to stronger multilingual contextual embeddings, which better capture semantic relationships in morphologically rich languages.

DistilBERT achieved moderate performance for Tamil, while SinBERT showed lower performance for Sinhala, likely due to dataset complexity and linguistic variability in emotional expression. Nevertheless, all models demonstrated the feasibility of emotion classification in low-resource settings, supporting integration with the emotion-aware text-to-speech pipeline.

Tamil emotion classification achieved strong results for Happy, Fear, and Surprise, which often exhibit clearer cues in social-media-style text. Lower performance for Anger and Sadness likely reflects lexical overlap among negative emotion categories. As shown in Tables I, II, and III, the overall results

TABLE III
SINHALA EMOTION CLASSIFICATION PERFORMANCE

| Emotion | Precision | Recall | F1-Score |
|---|---|---|---|
| Angry | 0.84 | 0.81 | 0.82 |
| Fear | 0.88 | 0.86 | 0.87 |
| Happy | 0.92 | 0.91 | 0.91 |
| Sad | 0.85 | 0.87 | 0.86 |
| Surprise | 0.89 | 0.90 | 0.89 |
| Neutral | 0.86 | 0.88 | 0.87 |

indicate robust multilingual emotion recognition performance. Overall Tamil results: Accuracy = 85.43%, Macro F1 = 0.83.

The Sinhala classifier showed strong performance across categories, especially Happy and Surprise, which have distinct markers in Sinhala usage. Slightly lower scores for Angry and Sad are consistent with contextual overlap in informal language. Overall Sinhala results: Accuracy = 87%, Macro F1 = 0.87.

### B. Acoustic Feature Analysis Results of the Sinhala Emotional Speech Dataset

To analyze the acoustic characteristics of the collected Sinhala emotional speech dataset, several audio features were extracted using the Librosa audio processing framework, including Root Mean Square (RMS) energy, Chroma features, Mel-spectrogram, and Mel-Frequency Cepstral Coefficients (MFCC). These features capture variations in energy, pitch distribution, spectral patterns, and timbral characteristics of emotional speech signals.

To ensure annotation reliability, inter-annotator agreement was evaluated using Fleiss' Kappa, which yielded a score of 0.74, indicating substantial agreement among annotators in labeling emotional speech categories.

Acoustic feature representations extracted from the emotional speech recordings illustrate that The Mel-spectrogram and MFCC representations highlight clear spectral variations across emotional states, while RMS energy and Chroma features capture differences in speech intensity and pitch distribution. These variations demonstrate that emotional states introduce measurable acoustic changes in the speech signal, which can be effectively utilized for emotion classification and expressive speech synthesis.

### C. Camera-Guided Document Capture Results

To validate the proposed camera-guided document capture module, all candidate models were trained on an NVIDIA RTX 3060 using 5-fold cross-validation and deployed on a mid-range Android device using PyTorch Lite for edge inference. Table IV summarizes mean Intersection over Union (mIoU), pixel accuracy, F1 score, inference latency, and model size.

BiSeNetV2 was selected as the best edge-deployment candidate, achieving 2.8 ms inference while maintaining strong segmentation accuracy (97.16% mIoU). Its compact size (7.6 MB) and low memory footprint support real-time 30 FPS guidance on resource-constrained devices, which is critical for continuous on-device camera feedback without cloud dependency. The

TABLE IV
CAMERA-GUIDED DOCUMENT CAPTURE MODEL COMPARISON

| Model | mIoU (%) | Pixel Acc (%) | F1 Score | Speed (ms) | Size (MB) |
|---|---|---|---|---|---|
| UNet + MobileNetV3 | 98.04 | 99.06 | 0.9905 | 5.7 | 23.0 |
| BiSeNetV2 | 97.16 | 98.65 | 0.9850 | 2.8 | 7.6 |
| SegFormer-B0 | 95.17 | 97.62 | 0.9730 | 5.8 | 14.3 |
| DeepLab + MobileNetV3 | 95.89 | 97.98 | 0.9795 | 4.6 | 42.1 |

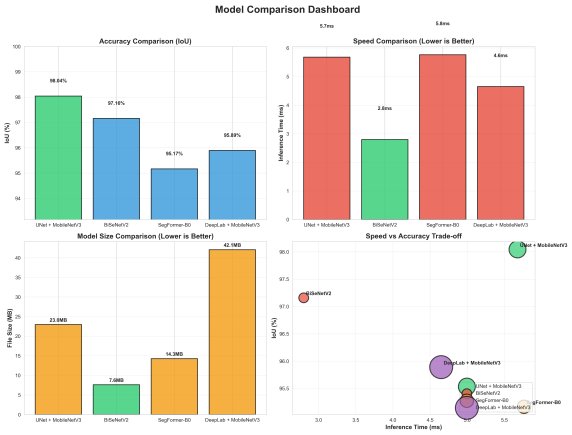


Fig. 1. Model Comparison Dashboard

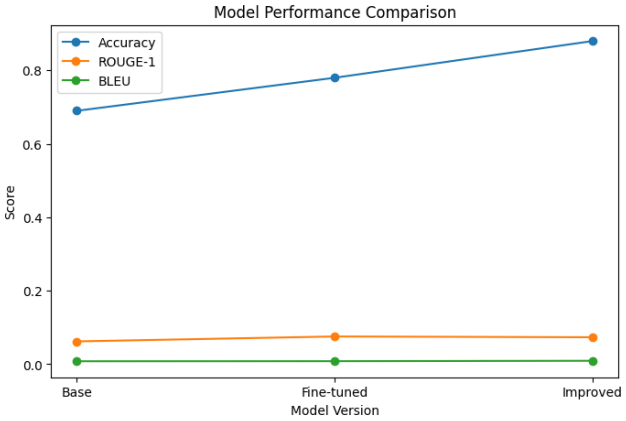


Fig. 2. Qualitative Tamil question-answering output from the fine-tuned model.

overall comparative behavior across candidate architectures is visualized in the Model Comparison Dashboard (Fig. 1), which complements Table IV by providing an at-a-glance view of the accuracy–latency trade-off.

### *D. Tamil Question Answering Module Results*

The experiments were conducted using the base model Qwen2.5-3B-Instruct. To adapt the model for Tamil question answering, parameter-efficient fine-tuning was performed using the Low-Rank Adaptation (LoRA) technique. Additionally, a reinforcement-learning recommendation module based on an epsilon-greedy contextual bandit tracks user reading behavior and updates category preferences using feedback to personalize news recommendations. Training and evaluation were carried out using the CHAI Tamil Question Answering dataset [13], which contains Tamil context passages, questions, and corresponding answers. The dataset was divided into training and testing subsets, and model performance was evaluated on a held-out test set.

Performance was assessed using widely adopted natural language generation metrics, including Accuracy, ROUGE [9], BLEU [10], and Exact Match. Table V reports the comparative performance across the base model, LoRA fine-tuned model, and improved training configuration, while Fig. 2 presents a qualitative Tamil QA output sample.

The results indicate that LoRA-based fine-tuning significantly improves the performance of the base model for Tamil question answering tasks. Accuracy increased from 0.69 in the base model to 0.78 after LoRA fine-tuning, and further improved to 0.88 with the enhanced training configuration. ROUGE scores also increased after fine-tuning, suggesting better lexical overlap between generated and reference answers. Additionally, the BLEU score showed gradual improvement, indicating stronger n-gram similarity between predicted and ground-truth responses.

The Exact Match score remained zero, which is common in generative question answering systems, as models tend to produce paraphrased or explanatory answers rather than exact string matches with the reference answers.

Overall, the results demonstrate that parameter-efficient fine-tuning with LoRA effectively enhances the capability of the Qwen2.5-3B model for Tamil question answering in low-resource language settings.

### *E. Sinhala Voice Command Intent Classification Module Results*

The Sinhala Voice Command Intent Classification Module is responsible for interpreting user voice commands and mapping them to predefined document interaction intents such as navigation, summarization, and question answering. The module processes the transcribed speech input and identifies the corresponding intent to determine the appropriate system action.

To evaluate the effectiveness of this component, a Sinhala voice command dataset was constructed and divided using an 80/20 stratified train–test split. A Linear Support Vector Machine (SVM) classifier with TF-IDF character n-gram features was used to model the intent classification task.

The model achieved an overall accuracy of 98.39%, demonstrating strong performance in recognizing Sinhala voice

TABLE V
TAMIL QUESTION ANSWERING PERFORMANCE ON CHAI DATASET

| Metric | Base Model | LoRA Fine-Tuned Model | Improved Training |
|---|---|---|---|
| Accuracy | 0.69 | 0.78 | 0.88 |
| ROUGE-1 | 0.0620 | 0.0754 | 0.0733 |
| ROUGE-2 | 0.0189 | 0.0208 | 0.0195 |
| ROUGE-L | 0.0615 | 0.0758 | 0.0732 |
| ROUGE-Lsum | 0.0609 | 0.0748 | 0.0721 |
| BLEU | 0.00794 | 0.00815 | 0.00918 |
| Exact Match | 0.00 | 0.00 | 0.00 |

TABLE VI
SINHALA VOICE COMMAND INTENT CLASSIFICATION PERFORMANCE

| Metric | Score |
|---|---|
| Overall Accuracy | 98.39% |
| Macro Average Precision | 0.98 |
| Macro Average Recall | 0.98 |
| Macro Average F1 | 0.98 |
| Weighted Average F1 | 0.98 |

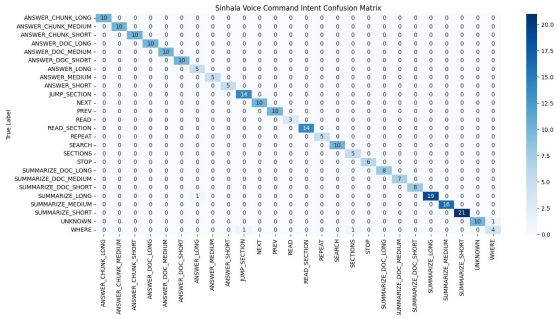


Fig. 3. Confusion matrix of Sinhala voice command intent classification.

commands across different interaction categories. Most intent classes achieved near-perfect precision, recall, and F1 scores, indicating reliable classification performance. As summarized in Table VI and visualized in Fig. 3, minor misclassifications were observed mainly between semantically similar intents such as WHERE and ANSWER_LONG.

These results demonstrate that the proposed intent classification model provides accurate and reliable interpretation of Sinhala voice commands, enabling effective voice-based interaction within the smart reader system.

## V. CONCLUSION

This research presented a comprehensive voice-agentic smart reader system designed to improve document accessibility for visually impaired users, with a particular focus on Sinhala and Tamil, two low-resource languages. The proposed system integrates four tightly coupled components: (i) structured Tamil article understanding with hybrid retrieval-based question answering, (ii) Sinhala voice-assisted non-linear navigation and command classification, (iii) emotion-aware text-to-speech synthesis for Sinhala and Tamil, and (iv) Intelligent Voice-Guided Navigation (IVGN) for non-visual document capture. At the document understanding level, the system introduces semantic topic segmentation, threshold-guided retrieval, constrained generation, and hybrid fallback mechanisms to improve reliability in Tamil question answering. These design choices demonstrate that structured retrieval control and segmentation are critical for reducing hallucination and improving contextual stability in low-resource language NLP systems. For interaction and navigation, the Sinhala voice-assisted command classification module enables natural, flexible, and non-linear document navigation using spoken Sinhala commands. By combining supervised machine learning, semantic retrieval using vector embeddings, and explicit navigation state management, the system moves beyond rigid linear screen-reader paradigms and supports intelligent content exploration. The emotion-aware TTS component further enhances accessibility by introducing affective awareness into speech output. Experimental results show that careful emotion classification, dataset curation, and controlled expressive synthesis significantly improve speech naturalness, engagement, and listening comfort compared to neutral narration. This is particularly important for visually impaired users who rely on prolonged auditory interaction. At the perception level, the IVGN component addresses the critical challenge of non-visual document capture. By transforming real-time semantic segmentation outputs into geometry-aware, prioritized auditory guidance, the system enables reliable OCR-ready image acquisition without visual feedback. Orientation-aware thresholds, temporal smoothing, and stability-based auto-capture collectively ensure cognitively manageable guidance and robust capture behavior. Taken together, these components demonstrate that accessibility-focused intelligent systems benefit more from structured design, explicit control mechanisms, and human-centered feedback than from model scale alone. The proposed framework shows that reliable, expressive, and interactive smart reading systems for low-resource languages are achievable through careful modular integration, even under limited data and computational constraints.

## VI. FUTURE WORK

Future work will prioritize stronger cross-module integration, including OCR-confidence-aware capture guidance, adaptive emotional TTS control by document context, and tighter coordination between semantic structure and voice navigation. Performance can be further improved through larger real-world datasets for Sinhala commands, Tamil QA, and emotional

speech, together with lightweight on-device Sinhala/Tamil language models optimized via parameter-efficient fine-tuning. Additional large-scale user studies with visually impaired participants are needed to measure long-term usability, listening fatigue, and robustness under challenging real-world conditions such as noise, motion blur, and low-light capture [20], [21], [22].

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